MACHINE LEARNING-BASED ANALYSIS OF HOUSEHOLD EXPENDITURE AND CONSUMPTION PATTERNS

A Study of Urban Households in Maharashtra

Machine Learning Project Report

Study Area: Maharashtra, India

ABSTRACT

This study applies machine learning techniques to examine household expenditure patterns among urban households in Maharashtra. The analysis focuses on monthly food and transportation expenditure. The dataset contains 10,567 household observations and includes income, district, age group, occupation, education, gender composition and household size. Six machine learning algorithms were evaluated using an 80:20 training-testing split. Their predictive performance was assessed using R-squared, Mean Absolute Error and Root Mean Squared Error. The best-performing models identified in the current evaluation were Linear Regression for transportation expenditure and Bagging Regression for food expenditure. Their test R-squared scores were 0.2542 and 0.5904, respectively. Permutation importance identified district as the most influential predictive feature for both expenditure categories, followed by household income. Household size also contributed substantially to food expenditure prediction.

1. INTRODUCTION

Household expenditure is an important indicator of economic well-being and consumption behaviour. Understanding how socioeconomic and demographic characteristics relate to household spending can help identify differences in consumption patterns. This project investigates food and transportation expenditure among urban households in Maharashtra using statistical analysis and machine learning techniques.

2. RESEARCH OBJECTIVES

1. Examine household food and transportation expenditure patterns.
2. Investigate relationships between income and expenditure.
3. Predict household expenditure using machine learning.
4. Compare six regression algorithms.
5. Identify important socioeconomic and demographic predictors.

3. DATASET DESCRIPTION

Parameter	Description
Total observations	10567
Geographical coverage	24 districts of Maharashtra
Region	Urban households
Numerical predictor	Total household income
Categorical predictors	District, age, occupation, education, gender, household size
Targets	Monthly food and transportation expenditure

4. METHODOLOGY

The analysis was conducted in several stages: 1. Data integration: Household income and expenditure datasets were merged using household ID. 2. Data cleaning: Missing values, duplicate records, data types and unusual observations were examined. 3. Exploratory analysis: Distributions and relationships between income and expenditure were investigated. 4. Statistical analysis: Descriptive statistics, Pearson correlation and Spearman correlation were calculated. 5. Machine learning preparation: Numerical and categorical predictors were identified. Categorical variables were one-hot encoded and numerical variables were standardised within the training pipeline. 6. Model evaluation: Six regression algorithms were compared using an 80:20 train-test split. 7. Feature importance: Permutation importance was calculated using the selected best-performing model for each target.

5. MACHINE LEARNING MODELS

The following algorithms were evaluated: 1. Linear Regression
2. Decision Tree Regressor
3. Random Forest Regressor
4. Gradient Boosting Regressor
5. Bootstrap Aggregation (Bagging Regressor)
6. Support Vector Regression

6. MODEL PERFORMANCE COMPARISON

Target	Model	R2	MAE	RMSE
Transportation	Linear Regression	0.2542	526.28	1098.36
Transportation	Decision Tree	-1.4132	631.65	1975.71
Transportation	Random Forest	-0.2379	585.99	1415.03
Transportation	Gradient Boosting	0.2231	529.07	1120.98
Transportation	Bagging Regressor	-0.1369	576.18	1356.11
Transportation	Support Vector Regression	0.1285	512.54	1187.30
Food	Linear Regression	0.5820	1863.88	2648.79
Food	Decision Tree	0.5189	1921.58	2841.51
Food	Random Forest	0.5812	1801.99	2651.17
Food	Gradient Boosting	0.5694	1882.47	2688.25
Food	Bagging Regressor	0.5904	1782.69	2621.88
Food	Support Vector Regression	0.2836	2377.20	3467.58

7. FEATURE IMPORTANCE ANALYSIS

Transportation Expenditure

Selected model: Linear Regression

DISTRICT: 0.34124

TOT_INCOME: 0.04325

SIZE_GROUP: 0.00282

EDU_GROUP: 0.00172

GENDER_GROUP: 0.00099

OCCUPATION_GROUP: 0.00091

AGE_GROUP: -0.00019

Food Expenditure

Selected model: Bagging Regressor

DISTRICT: 0.61869

TOT_INCOME: 0.18806

SIZE_GROUP: 0.15471

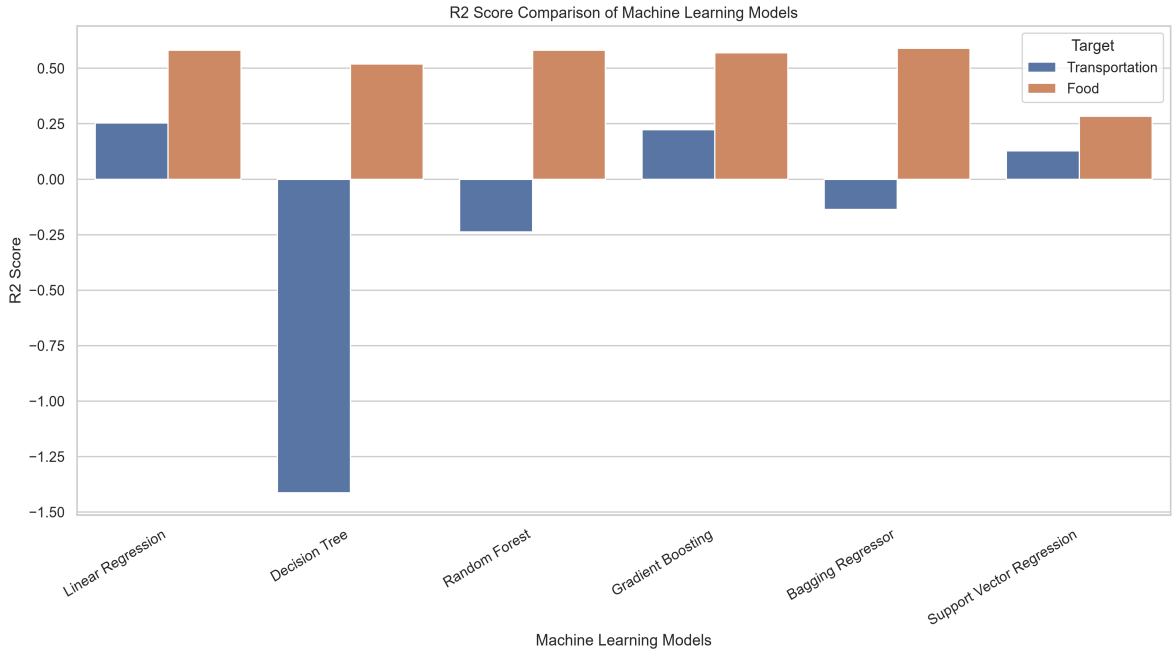
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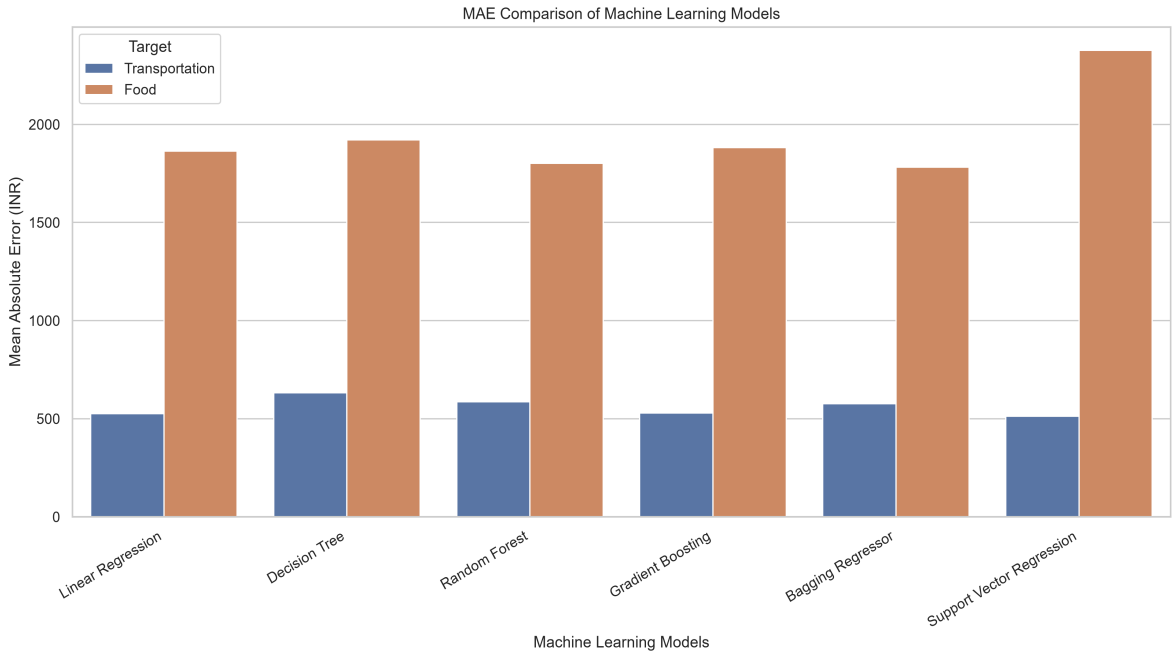
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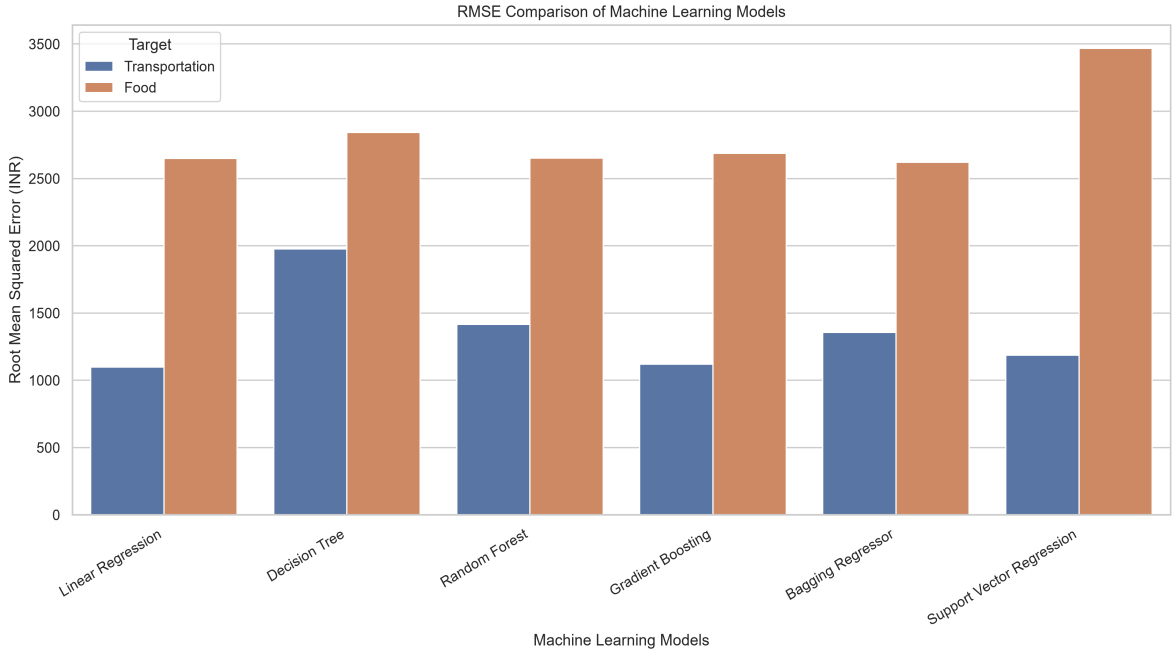
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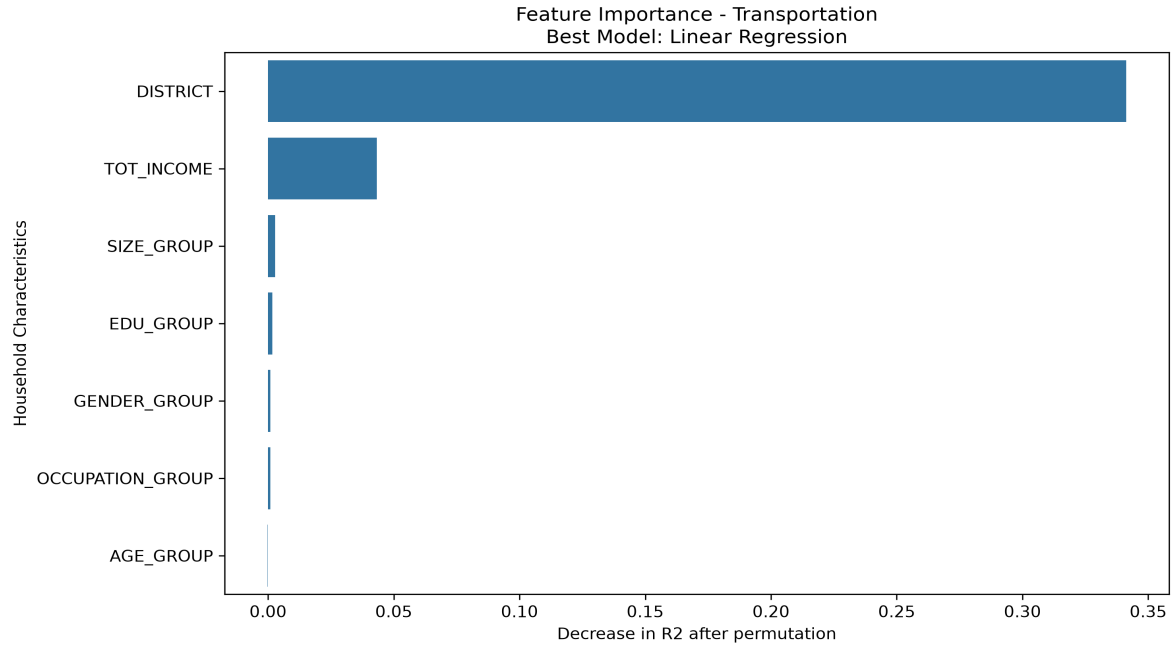
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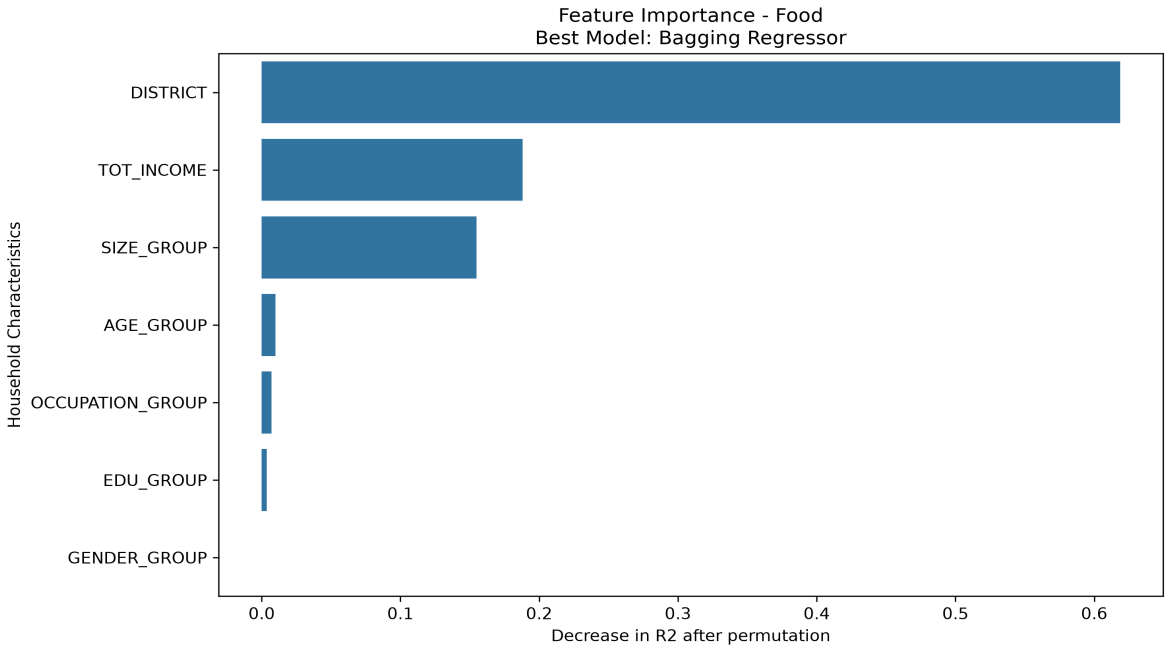
8. VISUAL RESULTS






RMSE Comparison of Machine Learning Models
Target
Transportation
Food
Root Mean Squared Error (INR)
3500
3000
2500
2000
1500
1000
500
0
Linear Regression
Decision Tree
Random Forest
Gradient Boosting
Bagging Regressor
Support Vector Regression
Machine Learning Models


Feature Importance - Transportation
Best Model: Linear Regression
Household Characteristics
DISTRICT
TOT_INCOME
SIZE_GROUP
EDU_GROUP
GENDER_GROUP
OCCUPATION_GROUP
AGE_GROUP
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
Decrease in R2 after permutation



9. RESULTS AND DISCUSSION

The transportation expenditure model achieved a test R-squared of 0.2542 using Linear Regression. This indicates comparatively limited predictive performance. The food expenditure model achieved a test R-squared of 0.5904 using Bagging Regression, indicating better predictive performance than the transportation model. District was the highest-ranked feature for both expenditure categories. Total household income ranked second in both cases. Household size was particularly relevant to food expenditure. These findings indicate that geographical and economic characteristics are important predictors in the evaluated models. However, predictive importance does not establish causality.

10. LIMITATIONS

1. The analysis is limited to the available urban household sample from Maharashtra. 2. Transportation expenditure has a highly right-skewed distribution. 3. The transportation prediction model has relatively low R-squared. 4. Important variables such as local prices, transport infrastructure, distance travelled and household consumption preferences are not included in the available dataset. 5. Model performance is based on a single train-test split. 6. Feature importance indicates predictive relevance rather than causal influence.

11. CONCLUSION

This project demonstrates the application of statistical methods and machine learning algorithms to household expenditure analysis in urban Maharashtra. Among the evaluated models, Linear Regression performed best for transportation expenditure, while Bagging Regression performed best for food expenditure. District emerged as the most important predictive feature for both targets. Income was also important, while household size had a substantial contribution to food expenditure prediction. The findings highlight differences in the predictability of household food and transportation spending and provide a foundation for more detailed socioeconomic consumption research.

12. FUTURE SCOPE

Future research can incorporate local price indices, fuel prices, public transport availability, household assets and detailed consumption characteristics. Additional improvements could include cross-validation, hyperparameter optimisation, target transformation for highly skewed expenditure variables and external validation using independent household datasets.

13. REFERENCES

1. Scikit-learn documentation: Machine Learning in Python.
2. Pandas documentation: Data analysis and manipulation.
3. NumPy documentation: Numerical computing in Python.
4. Matplotlib and Seaborn documentation: Data visualisation.
5. Original household income and expenditure Excel datasets provided for this project.